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Your Database Administrator Career Guide

Written by Coursera Staff • Updated on Oct 16, 2025
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Learn how to become a database administrator, including education, certification, and experience requirements.



More and more companies are using data management systems to inform decisions, produce products, and target specific audiences every day. Advances in business intelligence (BI) technology have improved data collection and analysis dramatically. The way companies interpret and store data is critical—which is where the role of a database administrator comes in.

Starting a career as a database administrator requires a combination of educational credentials, workplace and technical skills, practical experience, and continued professional development. It's a popular career choice with a higher-than-average salary, a promising job outlook, and opportunities for advancement. Learn more about how to become a database administrator, including job duties, skills, and career outlook.

What is the job outlook for database administrators?

According to the US Bureau of Labor Statistics (BLS), employment for database administrators is expected to grow 9 percent between 2023 and 2033, which is much faster than average [1]. This equates to around 9,500 positions per year [1]. Career progression is possible, with many database administrators becoming database architects.



What is a database administrator?

Database administrators are responsible for setting up the database structure and applications according to a company's needs. These needs typically include data storage, database documentation, database security, and system administration. This ensures the organization and accessibility of data for companies' strategic gain and the protection of sensitive data. Quick access to critical information is vital for customer service, compliance, and use in business development.



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What does a database administrator do?

The role of a database administrator varies according to the industry you work in and the type of data you're managing, but typically you'll have the following duties:

- Identifying user needs and appropriate database tools, such as database management software
- Considering the need for storage space, network requirements, and memory when designing database architecture
- Planning for database development, database installation, and continued database maintenance
- Updating database systems, associated servers, and applications
- Creating user profiles and managing access and permissions for database servers
- Completing regular functionality tests to measure database performance
- Creating database backups and recovery systems to ensure organizational data is secure
- Monitoring technical support of databases and documenting complex issues
- Using the database to produce queries for business intelligence reports

Database administrator job description

Database administration calls on a combination of workplace and technical skills to build and manage databases and to work with staff who need access. These include:

Technical skills

- Knowledge of computer operating systems
- Microsoft Access
- Oracle
- HTML
- SQL
- Linux
- Java
- Unix
- Sybase



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Cleansing, Plot (Graphics), Data Presentation, Regression Analysis, Descriptive Statistics, Scatter Plots, Predictive Modeling, Data Import/Export, Data Manipulation, Box Plots, Statistical Methods, Model Optimization, Statistical Analysis, Data Transformation, Model Training, Matplotlib, Seaborn, Histogram, Datamaps, Graphing, Data Collection, GitHub, Pandas (Python Package), Jupyter, Analytical Skills

Workplace skills

- Attention to detail
- Ability to prioritize
- Logical thinking
- Detail-oriented thinking
- Problem-solving abilities
- Organization
- Excellent communication skills
- Analytical approach

Database administrator roles

Many database administrators work in-house, looking after internal databases for a specific company. However, specialties also exist within database administration, breaking the role down into areas focused on by different people. These generally fall into three categories:

1. Database design and creation

The first specialty involves creating and designing databases to suit the user's needs. This includes using SQL, coding, and tuning. These people usually work for a technology or software development company as part of a team that can create larger systems of which the database is a part. Relevant job titles include database specialist and database analyst.

2. Database monitoring and system administration

The second specialty is performance database administration, which concerns optimizing the performance of databases and making tweaks to improve performance. The role is proactive because it focuses on improving performance to avoid problems. Relevant job titles include database administrator, systems administrator, and management information systems (MIS) administrator.

3. Database applications and environment

The last specialty is application database administration, which concentrates on integrating applications into databases. This back-end work requires the use of a query language such as SQL. Application database

administrators also write and debug programs as well as maintain applications. Relevant job titles include database programmer, database software developer, and database engineer.



What is it like working as a database administrator?

Database administration spans industries, so work in this field comes with many choices. You may work in retail, health care, hospitality, or research, to name a few. According to US News & World Report, database administrator ranked as the eighth-best technology job [2] and comes with a good salary, career advancement, and work-life balance.

Where do database administrators work?

The role of a database administrator is usually a full-time, office-based position, often as part of an IT team. Being a database administrator can be rewarding but also challenging, with a high responsibility for sensitive data and a need to follow data protection laws. US News & World Report scores database administration a 3.8 out of 10 for work-life balance and a 4.5 for stress [2]. As the career is ever-changing as technology evolves, the role can be exciting, and being adaptable is essential.

How much does a database administrator make?

According to the BLS, the median annual salary for a database adminstrator is $101,510 [3], which is above average. The top 10 percent in the role earn more than $157,710 per year [3].

How to become a database administrator

Requirements for these roles vary depending on industry, company, and location. 61 percent of database administrators have a bachelor's degree, 15 percent have an associate degree, and 13 percent have a master's degree [4]. Relevant areas of study include business, computer science, accounting, and computer information systems. Demonstrating your ability to use specific software and having technical skills beyond

what's taught in a degree are essential. Many companies ask for training or certification in the database software they use in their organization.

Gain IT experience.

Database administrators must demonstrate specific technical skills in addition to holding a degree and certifications. These skills can be from a previous position, such as an entry-level job like data entry or another IT role. You'll need experience to start your career before you can work up to managing sensitive data.

Earn database administrator certifications.

The value employers place on database administrator certifications is high. Companies require more than a degree and experience, as they want to know that you have relevant database management skills and can use the appropriate database languages. Certification in the right software, management system, or language can set you apart, even if it isn't a requirement.

Employers sometimes provide on-the-job training and expect database administrators to engage in professional development. Consider the following certifications and online training you may want to pursue:

- Oracle Certified Professional (OCP)
- IBM Data Science Fundamentals with Python and SQL Specialization
- Oracle Autonomous Database Professional Workshop
- Computer Service Technician (CST)
- IBM Data Analyst Professional Certificate
- Microsoft Certified: Azure Database Administrator Associate
- Google Data Analytics Professional Certificate
- IBM Certified Database Associate - DB2 9 Fundamentals
- Meta Introduction to Databases



Take the next step in your database administrator career on Coursera

As a database administrator, you can work with database management systems and even prepare for a future role as a database manager. If you're looking to begin your career as a database administrator, starting with some courses and certificates is a great first step. You can develop a foundation in database management with the IBM Data Science Professional Certificate or Database Management Essentials from the University of Colorado on Coursera to launch your database management career.



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Article sources

1.

US Bureau of Labor Statistics. “Occupational Outlook Handbook: Database Administrators and Architects: Summary, https://www.bls.gov/ooh/computer-and-information-technology/database-administrators.htm#tab-1.” Accessed February 4, 2025.

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